WIZM Yellow Paper

State Transition, PoA Consensus, Fees, and Transaction Layouts

WIZM Protocol Engineering

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Abstract

This document specifies the current WIZM chain: a five-validator proof-of-authority ledger with transparent and stealth transfers. The chain uses Falcon-512 signatures, ML-KEM for encrypted mempool and transport key establishment, BLAKE3 hashing, and AES-256-GCM mempool encryption. It contains no zero-knowledge, shielded, confidential, or note-based transaction path. Stealth transfers provide recipient unlinkability only; sender and amount are public.

1 Network parameters

Parameter	Value
Chain ID	7777
Validators	5 PoA seats
Steady slot	10 seconds
Genesis slot	240 seconds
Slow-start slot	60 seconds
Max transactions per block	1200
Max block body	8 MiB
Max sealed throughput	120 TPS
Base fee	10,000 base units
Burst-free transactions per wallet/block	20
Burst multiplier	10x per excess transaction

The proposal cap is the count cap first for transparent and stealth transactions. A block is invalid if it exceeds 1200 transactions or 8 MiB, or if a wallet exceeds the burst schedule.

2 Cryptographic functions

Let $\mathrm{BLAKE3} : \{0,1\}^* \to \{0,1\}^{256}$ be BLAKE3-256. An address is

$$A(pk) = \mathrm{BLAKE3}(pk)[0:20] \in \{0,1\}^{160}.$$

Falcon-512 provides signatures:

$$\mathrm{Verify}_{pk}(m, \sigma) = 1.$$

Public keys are 897 bytes; signatures are 512–752 bytes. ML-KEM-512 is used for encrypted mempool/committee key establishment; ML-KEM-768 is used for transport. AES-256-GCM encrypts mempool payloads.

3 State

A state σ maps a 20-byte address to

$$\sigma(a) = (balance, nonce, public_key).$$

It also contains spent stealth key images, block metadata, committee state, and tokenomics counters. There is no shielded commitment tree, note set, nullifier set, or Merkle proof state.

For a sender a and amount v with fee f, a transparent transfer is valid only if

$$balance(a) \geq v + f, \qquad nonce(tx) = nonce(a).$$

After execution:

$$balance(a) \leftarrow balance(a) - v - f, \quad nonce(a) \leftarrow nonce(a) + 1.$$

4 Transaction layouts

All integers are little-endian. Falcon signatures are variable-length.

4.1 Transparent transfer

$$[from : 20] \parallel [to : 20] \parallel [amount : 8] \parallel [nonce : 8] \parallel [pk : 897] \parallel [sig : 512..752].$$

Wire size is 1,465–1,705 bytes. Sender, recipient, and amount are public. The signature covers the first 56 bytes.

4.2 Stealth transfer

$[sender : 20]\|[stealth : 20]\|[ephemeral : 32]\|[view_tag : 1]\|[amount : 8]\|[key_image : 32]\|[pk : 897]\|[sig : 512.$

Wire size is 1,522–1,762 bytes. The 113-byte body is signed. Sender and amount are public. The one-time `stealth` address is derived from the recipient's spend key and an ephemeral shared value. The key image prevents a second spend. There is no encrypted amount and no zero-knowledge proof.

The address derivation is

$$shared = \text{BLAKE3}(ephemeral),$$

$$stealth = \text{BLAKE3}(spend_pk \parallel shared)[0:20],$$

$$view_tag = shared[0].$$

The recipient scans with the view secret and recomputes the expected address. The sender is visible in the transaction.

5 PoA consensus

Validators take turns producing blocks in ten-second slots. A producer drains at most 1200 transactions from the encrypted mempool, seals a block, and broadcasts it. Validators verify the block and sign a notarization vote over the chain/genesis domain, block hash, and height. A majority of active seats advances finality. This is a crash-fault finality rule, not a Byzantine-fault-tolerant consensus proof.

Competing blocks at the same height are resolved deterministically: the earlier seal wins, with a deterministic tie-break and a bounded unwind. Nodes synchronize full block bodies through unicast and gossip; they update their mesh-best view only from locally validated blocks.

6 Fees and tokenomics

Let n be the wallet's index among its fee-paying transactions in a block. The burst multiplier is

$$\mu(n) = \begin{cases} 1, & n \le 20, \\ 10, & n = 21, \\ 100, & n = 22, \\ 1000, & n = 23, \\ \dots \end{cases}$$

The charged fee is

$$f(n) = 10{,}000 \cdot \mu(n).$$

During emission, fees are burned. After emission, fees are distributed to validators. The block reward is

$$R_{validator} = \frac{E_{period}}{B_{period} \cdot N_{validators}}$$

where E_{period} is the emission for the current two-year period, $B_{period} = 6{,}311{,}520$ blocks, and $N_{validators} = 5$. At the current period this is 88,126,450 base units, or 0.8812645 WIZM per validator per block.

7 Capacity

For a block with T transactions and body size S:

$$T \le 1200, \qquad S \le 8 \cdot 1024^2.$$

At a ten-second slot,

$$\Theta_{\max} = \frac{1200}{10} = 120 \text{ TPS}.$$

A generator may submit faster than the chain seals; those transactions remain in the encrypted mempool. The chain itself never seals more than 120 TPS under the current cap. The burst-free wallet limit is 20 transactions per block, so at least 60 distinct wallets are needed to fill a 1,200-transaction block without multiplied fees.

8 What was removed

The following are not part of the current protocol:

- shielded transfers, shielded deposits, and unshields;
- notes, commitments, nullifiers, and the shielded Merkle tree;
- zero-knowledge proofs and proof verification;
- encrypted stealth amounts;
- shielded wallet, RPC, CLI, and Tauri surfaces.

Stealth is not confidential. It provides one-time recipient addresses and key-image double-spend protection only. Sender and amount are public consensus data.

9 Verification obligations

A conforming implementation must reject:

- any transaction outside the 1,200/8 MiB block limit;
- any stealth transaction outside 1,522–1,762 bytes;
- any wallet above the 20-transaction burst-free window without the multiplied fee;
- any replay or duplicate key image;
- any attempt to introduce removed shielded/ZK transaction types.

The test suite and load harness must exercise only transparent and stealth transactions.